

Valeria Ramirez Osorio

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EDUCATION

McMaster University

Computing and Software

University of Toronto

Computer Science Specialist and Mathematics Minor

- 3.6 CGPA on a 4.0 Scale
- Dean's List (2021, 2022, 2023, 2024)
- Ontario Scholar Certificate (2020)

Masters of Engineering

June 2027

Honors Bachelor of Science

Expected July 2025

SKILLS HIGHLIGHTS

Software Skills

- Python, Java, JavaScript, NodeJS, React, C, C++, Bash, HTML and CSS, UNIX/Linux, Microsoft Products.
- Web and Application Development, Frontend and Backend, Full Stack, Automation, Model Evaluation and Optimization, High Critical Thinking Skills, Object-Oriented Design, Complexity Analysis, Algorithms

Data Science Skills

- MySQL, PostgreSQL, R, Microsoft Excel, Google Cloud Platform, Docker, Kubernetes
- Data Interpretation and Manipulation, Neural Networks, Numerical analytics, Probability and Statistics

Spoken Language Skills

- English (Fluent), Spanish (Fluent), French (Intermediate)

PRIMARY EXPERIENCE

Summer Student Engineering Internship

May 2021 – August 2024

Bell Mobility

- Led annual projects aimed at enhancing team efficiency, including application development, inventory database implementation, and software testing/debugging through Python, Java, JavaScript, HTML, SQL, etc.
- Contributed to team success, such as 5G Slicing, RAN, and SRE, through collaboration, effective communication, in Agile and Scrum practices. Used Jira and Confluence to manage the SDLC and ensure timely project delivery.
- Applied analytical and problem-solving skills, adaptability, and initiative to provide innovative solutions, manage priorities, and support a productive team environment.

Application and Database Developer

September 2024 – Present

University of Toronto

- Independently created and implemented a website's front-end and back-end, leveraging full creative freedom to achieve a unique design and functionality to implement into the Department of the Historical Studies.
- Utilized SQL, Oracle, Omeka, React, Node.js, JavaScript, HTML, and other tools to develop the platform.

PROJECTS

Development of an AI-Driven Chatbot | *Master's Research Project*

September 2025 – Present

- Designing and developing an interactive chatbot application powered by generative AI to simulate professional client interactions and enhance engineering students' requirement elicitation skills.
- Implementing frontend architecture and user experience design to create realistic, classroom-applicable client communication scenarios.
- Collaborating with a research supervisor and graduate peers to iterate towards a functional prototype and prepare for a formal user study evaluating educational impact.

Understanding the Impact of Using Generative AI Tools in a Database Course | May 2024 – August 2024

- Manipulated and organized quantitative and qualitative data using Python and Microsoft Excel to extend trends for computer science professor to understand interaction of students and GenAI in database courses.
- Collaborated with two professors and a graduate student to construct a seven page research paper to be published by SIGCSE for data scientists and professionals at the SIGCSE '25 conference.

Multiple Representations in Introductory Programming: A Pilot Study |

September 2024 – Present

- Developed the backend programming, using Javascript, of a series of instructional programs for undergraduate students to conceptually understand python to be implemented into first year courses at the University of Toronto.

ADDITIONAL EXPERIENCE

Teaching Assistant | *University of Toronto*

September 2022 – Present

- Facilitated engaging tutorials, supported lecture delivery, graded assessments, invigilated exams, and provided individualized academic assistance through office hours to enhance student learning outcomes.
- Courses Include: Intro to Databases, Intro to Computational Theory, and Computer Organization

Computer Science and Mathematics Tutor | *Self-employed*

September 2024 – Present

- Crafted lessons and practices for high school students to improve their understanding and knowledge.
- Focused on Advanced Functions, Calculus, Computer Science, Languages: Java and Python
- Experienced in supporting students with diverse learning needs, including those with dyslexia, dyscalculia, and ADHD.